

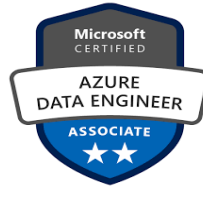
Anjali Erra

(Data Engineer)

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PROFESSIONAL SUMMARY

- Having around 5+ years of experience in software development with a focus on **Big Data, Azure Cloud Engineering, Data Warehousing**, and the **Hadoop Ecosystem**.
- Hands-on experience with **Azure** services such as **Azure HDInsight, Azure Data Lake, Azure Synapse Analytics, Data Factory, Azure Blob Storage, Cosmos DB**, and **Stream Analytics** for efficient data processing and storage.
- Utilized **Azure Virtual Machines (VMs), Azure Blob Storage, Azure Data Lake**, and **Azure Kubernetes Service (AKS)** for resource management, data orchestration, and real-time processing.
- Employed various **AWS** services, including **Amazon EC2, S3, EMR, RDS, VPC, Elastic Load Balancing, IAM, Auto Scaling, CloudFront, CloudWatch**, and **Lambda** for efficient resource management and automation.
- Deep expertise with **Hadoop** tools and services across **Azure HDInsight** and **Hortonworks**, including **HDFS, YARN, Hive**, and **Pig** for large-scale data processing.
- Managed terabytes of streaming data using **Kafka, Azure Event Hubs, Spark Streaming**, and **Storm**, while orchestrating workflows with **Azure Data Factory** and **Apache Airflow**.
- Developed production-ready **Spark**-based applications with components like **Spark SQL, Data Frames, Datasets**, and **Spark Streaming** for optimized processing of real-time data.
- Implemented **AWS Lambda** functions for serverless data processing and automation, triggering workflows and ETL pipelines based on specific events in **S3** and **DynamoDB**.
- Utilized **Microsoft Fabric** unified platform to build end-to-end analytics solutions seamlessly integrating data engineering, data science, and business intelligence.
- Designed and implemented **data lake architectures** using **AWS Lake Formation** and **AWS Glue**, ensuring secure and efficient data cataloging, transformation, and governance.
- Built and maintained **data pipelines** using **AWS Glue, Step Functions**, and **Lambda**, automating ETL workflows and reducing manual intervention.
- Designed and implemented **Kafka** Producers and Consumers on **Azure** for real-time data ingestion and processing, storing data in **Azure Data Lake** and analyzing it using **Azure Synapse**.
- Proficient in both **SQL** and **NoSQL** databases such as **Azure SQL Database, Cosmos DB**, and **MongoDB**, with a strong focus on data modeling, query optimization, and disaster recovery.
- Expertise in scripting using **Python (PySpark), Scala**, and **Spark SQL** for efficient **data aggregation** and ETL processes across various data formats like **JSON, CSV**, and **Parquet**.
- Worked with Data Factory pipelines in **Microsoft Fabric** for integrating data from diverse sources into a unified platform.
- Utilized **AWS Kinesis** for real-time data streaming, enabling the ingestion and analysis of high-velocity data in real-time from multiple sources.
- Used **power Query** for Incremental Refresh to Optimize data refresh performance in Power BI Premium and Azure services.
- Orchestrated data workflows using **Apache Airflow** for scheduling and automating daily data ingestion.
- Integrated data from **on-prem** and cloud sources into **Redshift**, processing huge volumes in real-time using **AWS Glue**.
- Executed ETL pipelines using **Azure Data Factory** and processed data using **U-SQL** and **Azure Synapse Analytics** to derive actionable insights
- Managed large **Cosmos DB clusters**, monitoring document growth, performing indexing, replication, load balancing, and implementing ad-hoc queries to improve performance.
- Built **Azure Kubernetes Service (AKS)** clusters using **ARM** templates and **Azure CLI** for automating cloud environment deployments.

- Used **DAX (Data Analysis Expressions)** in **Power BI, Power Pivot, and Azure Analysis Services** for advanced calculations.
- Developed scalable **REST APIs** and **ETL** solutions using Informatica on **Azure**, ensuring high performance and reliability.
- Used **power query** to Work with **Power Automate** for **data pipeline automation**.
- Utilized **Git, DevOps**, and **JIRA** for version control, bug tracking, and seamless collaboration in the development lifecycle.
- Participated in all phases of the **SDLC** using both **Waterfall** and **Agile Scrum** methodologies, ensuring smooth project execution and timely deliveries.
- Migrated legacy applications to **Cloud** using **DevOps** tools like **GitHub, Jenkins, Docker**, and **Terraform** for efficient automation and deployment.
- Collaborated effectively with cross-functional teams, including business stakeholders, production support, and engineering teams to support data analysis and decision-making on analytics platforms.

CERTIFICATIONS

- Certified Azure Data Engineer Associate (DP 203)
- Certified Azure Fundamentals(AZ-900)
- Certified Azure Administrator Associate(AZ-104)
- Certified AWS Data Engineer Associate(DEA-C01)

TECHNICAL SKILLS

Programming Languages	C, J2EE, SQL, Scala, Python, Unix Shell Scripting.
Databases	Azure SQL Database, Cosmos DB, MySQL, PostgreSQL, Teradata, Oracle, MS-SQL Server, RDBMS, DB2, AWS S3
Big Data Technologies:	HDFS, Yarn, MapReduce, Hive, HBase, Cassandra, Apache Spark, Kafka, Azure HDInsight, Azure Databricks
Cloud	Azure HDInsight (Databricks, Data Lake, Blob Storage, Data Factory, SQL DB, SQL DWH, Cosmos DB, Azure DevOps, Active Directory, Fabric) , AWS (EMR, EC2, EBS, RDS, S3, Glue, Elasticsearch, Lambda, Kinesis, SQS, DynamoDB, Redshift, ECS)
NoSQL Database	Cassandra, MongoDB.
Scheduling tools/ETL	SQL Server Agent, Visual Studio, Databricks, Data Factory, Synapse Analytics
Reporting tools	Informatica, Talend, SSIS, SSRS, SSAS, Tableau, Power BI.
Source Control	Git, CVS, SVN, Perforce, GitHub and Microsoft Azure DevOps (TFS)
Methodologies:	Agile/Scrum, Waterfall.
Development Tools	Eclipse, NetBeans, IntelliJ, Microsoft Office Suite (Word, Excel, PowerPoint, Access)
Others	Stream Sets, Spring Boot, Jupyter Notebook, Docker, Kubernetes, Jenkins, Jira.

WORK EXPERIENCE

AWS Data Engineer | Azure Data Engineer
State of Nebraska

Nov 2024 – Present

- Enhanced query performance on **Azure SQL Database** and **Azure Synapse Analytics** by implementing advanced query tuning techniques, improving database and data warehouse efficiency.

- Utilized **Azure Data Lake Analytics** and **Azure Synapse** to perform complex data analysis on large datasets stored in **Azure Blob Storage**, leveraging **U-SQL** and **Spark SQL** for optimized **ETL** processes.
- Designed and implemented **Python**-based solutions to extract real-time data from **Azure Blob Storage**, integrating it into **Azure SQL Database** for enhanced reporting and analytics capabilities.
- Automated the deployment of applications to **Azure Service Fabric clusters** using **CI/CD** pipelines integrated with **Azure DevOps**.
- Developed scalable, **object-oriented** solutions using **Python** and **Java** for large-scale data processing and application development, ensuring optimal performance and maintainability.
- Worked with **Azure Databricks** to manage a **Delta Lake environment**, utilizing **Apache Spark** for processing and analyzing high-volume data streams in real-time scenarios.
- Used **Power Query** as a **self-service ETL (Extract, Transform, Load)** tool used in **Power BI, Excel, and Power Platform**.
- Executed large-scale **ETL pipelines** using **Azure Data Factory, Spark SQL, and Azure Synapse Analytics**, processing data from **Blob Storage** to deliver real-time insights for business decision-making.
- Processed terabytes of data daily using **AWS** services like **EMR, Redshift, S3, and Glue** for real-time data ingestion, transformation, and analysis.
- Automated data pipelines using **Spark, AWS Glue, and Step Functions** to pull data from relational **databases (RDBMS)** into **S3**, ensuring seamless daily data updates.
- Built workflows to load incremental data daily, moving from traditional **RDBMS** to **AWS data lakes** for faster data access and real-time insights.
- Executed **ETL** jobs to transfer data from various sources to **S3** and **Redshift** using **Glue, Spark SQL, and Athena**, ensuring efficient, real-time data processing.
- Developed **ETL** pipelines to handle large-scale data integration between **RDS, S3, and Redshift**, processing millions of records in real-time.
- Orchestrated data workflows using **Apache Airflow** for scheduling and automating daily data ingestion.
- Integrated data from **on-prem** and cloud sources into **Redshift**, processing huge volumes in real-time using **AWS Glue**.
- Optimized **SQL** scripts and used **PySpark** to improve the performance of real-time data pipelines.
- Processed complex **ETL** tasks with **Spark** and **AWS Glue**, enabling real-time data analysis.
- Built data models using **star** and **snowflake** schema, improving real-time query performance on large datasets.
- Used **Glue** and **EMR** notebooks for quick, interactive analytics on large datasets.

Environments: Azure SQL Database, SSMS, Azure Synapse Analytics, U-SQL / Spark SQL, Azure Databricks, Python, Azure Fabric, Azure DevOps (Repos, Pipelines), AWS, Amazon RDS, Amazon Redshift, AWS S3, AWS Glue, AWS EMR, Apache Spark, AWS Lambda, AWS Kinesis, AWS Step Functions, AWS Athena, AWS Code Pipeline, GitHub / GitHub Actions, CI/CD, Python, Java, Kafka, Azure Functions, Jenkins, Azure Blob Storage, Azure Event Hubs, Apache Airflow, Azure Data Factory, Azure Synapse Analytics, Spark SQL, Azure Databricks, ADLS Gen2

Azure Data Engineer

May 2023 – Nov 2024

Johnson & Johnson

- Automated the deployment of applications to **Azure Service Fabric clusters** using **CI/CD** pipelines integrated with **Azure DevOps**.
- Developed scalable, **object-oriented** solutions using **Python** and **Java** for large-scale data processing and application development, ensuring optimal performance and maintainability.
- Worked extensively with **SAP Data Services Designer** for ETL processes, ensuring efficient data extraction, transformation, and loading.
- Gained hands-on experience with **SAP BI Portal, Web Intelligence (WebI), and Information Design Tool (IDT)** to develop and manage enterprise-level reports and dashboards for business insights.
- Worked with **Azure Databricks** to manage a **Delta Lake environment**, utilizing **Apache Spark** for processing and analyzing high-volume data streams in real-time scenarios.
- Design, develop, and optimize ETL/ELT pipelines for ingesting, transforming, and loading data into target systems.
- Implement data validation, cleansing, and transformation logic using **SQL, Python, or R**.
- Automate data workflows using **Azure Data Factory (ADF)** or other cloud-based tools.

- Create and manage **SQL Server tables, views, stored procedures, and functions** to support actuarial processes.
- Optimize database queries for performance improvements and ensure efficient data retrieval.
- Develop and maintain **SSIS (SQL Server Integration Services) packages** for data movement and transformation.
- Implement data quality checks and enforce data governance policies.
- Utilized **Azure Data Lake Analytics** and **Azure Synapse** to perform complex data analysis on large datasets stored in **Azure Blob Storage**, leveraging **U-SQL** and **Spark SQL** for optimized **ETL** processes.
- Designed and implemented **Python**-based solutions to extract real-time data from **Azure Blob Storage**, integrating it into **Azure SQL Database** for enhanced reporting and analytics capabilities.
- Automated the deployment of applications to **Azure Service Fabric clusters** using **CI/CD** pipelines integrated with **Azure DevOps**.
- Developed scalable, **object-oriented** solutions using **Python** and **Java** for large-scale data processing and application development, ensuring optimal performance and maintainability.
- Worked with **Azure Databricks** to manage a **Delta Lake environment**, utilizing **Apache Spark** for processing and analyzing high-volume data streams in real-time scenarios.
- Used **Power Query** as a **self-service ETL (Extract, Transform, Load) tool** used in **Power BI, Excel, and Power Platform**.

Environments: Azure Service Fabric, Azure DevOps, Python, Azure Databricks, Delta Lake, Apache Spark, SQL Server, SSIS, Power BI, SSRS, Azure Data Factory (ADF), Azure SQL Database, Azure Synapse Analytics, Azure Data Lake Storage (ADLS), U-SQL, Spark SQL

AWS Data Engineer **Harman Connected Services, Bangalore, India**

Aug 2021 – July 2022

- Optimized query performance on **AWS** services like Amazon **RDS** and Redshift, increasing database and data warehouse efficiency by 32%. Utilized advanced indexing and query-tuning techniques for faster, real-time data retrieval.
- Leveraged **AWS Athena** to analyze and manipulate large volumes of CSV data stored in **S3** using **Scala** queries for real-time data extraction, transformation, and loading (**ETL**) processes.
- Developed **Python**-based solutions to pull real-time data from **AWS S3**, feeding insights into **SQL Server** to meet business reporting and analytics needs.
- Conducted **ETL** processes on large data sets using **Python**, **Spark SQL**, **S3**, and Redshift to derive actionable customer insights in real time.
- Improved internal processes by automating manual tasks using **Python** and **Airflow**, managing real-time incremental loads, and optimizing data storage and delivery for scalable analytics.
- Utilized **Spark**'s in-memory computing to process extensive datasets **on S3 Data Lake** in real-time, filtering and loading data into **Hive** external tables for real-time analytics.
- Orchestrated automated **CI/CD** pipelines with **AWS Code Pipeline**, **Jenkins**, and **AWS Code Deploy**, improving real-time deployment efficiency.
- Authored **PySpark** scripts to streamline real-time **ETL** processes, utilizing crawlers to extract data from **S3** and generate a data catalog for metadata management across multiple data stores (**Amazon S3, DynamoDB**).
- Implemented **AWS Lambda** functions for serverless data processing and automation, triggering workflows and **ETL** pipelines based on specific events in **S3** and **DynamoDB**.
- Utilized **AWS Kinesis** for real-time data streaming, enabling the ingestion and analysis of high-velocity data in real-time from multiple sources.
- Set up and managed **Amazon Redshift** clusters, optimized queries, and configured distribution keys to enhance data warehouse performance for large-scale analytics.
- Designed and implemented **data lake architectures** using **AWS Lake Formation** and **AWS Glue**, ensuring secure and efficient data cataloging, transformation, and governance.
- Built and maintained **data pipelines** using **AWS Glue**, **Step Functions**, and **Lambda**, automating **ETL** workflows and reducing manual intervention.
- Implemented **data security** and **compliance measures**, including **IAM roles**, **encryption (S3 SSE-KMS)**, and **VPC configurations** to ensure data privacy and compliance with **GDPR, HIPAA, and SOC 2**.

- Optimized **Amazon Redshift performance** by using **materialized views, compression encodings, and workload management (WLM)** to improve query execution efficiency.
- Developed event-driven architectures using **AWS Lambda, SNS, and SQS**, enabling real-time data processing and notifications for business-critical applications.
- Automated schema evolution in **AWS Glue and Athena** by leveraging **Apache Iceberg and Delta Lake**, ensuring seamless handling of evolving data structures.

Environments: AWS, Amazon RDS, Amazon Redshift, AWS S3, AWS Glue, AWS EMR, Apache Spark, AWS Lambda, AWS Kinesis, AWS Step Functions, AWS Athena, AWS Code Pipeline, AWS Code Build, AWS Code Deploy, AWS Quick Sight, AWS DynamoDB, AWS Lake Formation, Apache Airflow, MongoDB, SQL Server, AWS Data Lake, AWS CloudFormation, AWS VPC, AWS IAM.

Data Engineer/ Data Analyst DXC Technologies, India

June 2019 - July 2021

- Utilized **Azure Data Factory** to build robust **ETL** pipelines for ingesting data from various sources into **Azure Data Lake Gen2**.
- Implemented data transformation processes using **Data Flow** features to ensure data quality and consistency.
- Used **Azure Databricks** for real-time data processing and analytics. Utilized **Apache Spark** to process streaming data, enabling immediate insights and enhancing decision-making.
- Designed and managed **Azure Data Lake Gen2** architectures for scalable storage solutions, facilitating the efficient organization and retrieval of large datasets for analytics.
- Employed **Azure Databricks** notebooks to promote collaboration between data engineers, enabling joint efforts in **data exploration, machine learning, and visualization** tasks.
- Implemented data quality checks and transformations in **Azure Data Factory** to ensure the integrity of data before it enters analytical processes.
- Created interactive dashboards and reports using **Power BI** to present insights from data, making complex data easy to understand for stakeholders and aiding in strategic decision-making.
- Conducted data exploration and analysis using **SQL** to query large datasets stored in **Azure SQL Database** and **Azure Data Lake**, extracting actionable insights for business teams.
- Integrated various **BI** tools such as **Looker** and **Tableau** with **Azure data sources** to enhance reporting capabilities and provide dynamic visualizations for end-users.

Environments: Looker, Tableau, Power BI, Azure Synapse Analytics, Spark SQL, Azure Databricks, ADLS Gen2, Azure SQL Database, SSMS, Azure Synapse Analytics, U-SQL / Spark SQL, Azure Databricks, Python, Azure Fabric, Azure DevOps (Repos, Pipelines), GitHub / GitHub Actions, Azure Functions, Jenkins, Azure Blob Storage, Azure Event Hubs, Apache Airflow, Azure Data Factory.

EDUCATION

Master's, Computer Science, University of Central Missouri, MO - 3.60/4.00